



WORK SHEET - 22

INDEFINITE & DEFINITE INTEGRATION

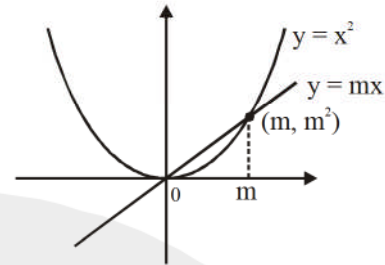
Q.1 (A)

$$A_m = \int_0^m (mx - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^m = \frac{m^3}{2} - \frac{m^3}{3}$$

$$\Rightarrow A_m = \frac{m^3}{6}$$

$$\Rightarrow \sum_{m=1}^n A_m = \frac{1}{24} n^2 (n+1)^2$$

$$\therefore \lim_{n \rightarrow \infty} \frac{\sum_{m=1}^n A_m}{n^4} = \lim_{n \rightarrow \infty} \frac{1}{24} \frac{n^2 (n+1)^2}{n^4} = \frac{1}{24} \text{ Ans.}$$



Q.2 (D)

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \left(\frac{r}{n} \right)^{2013} \times \frac{1}{n} = \int_0^1 x^{2013} dx = \frac{1}{2014} \text{ Ans.}$$

Q.3 (A)

$$f(x) = e^{x^3-3x} \text{ and } f(-x) = e^{-(x^3-3x)} = \frac{1}{f(x)} \quad [e^a > 0 \Rightarrow f(x) > 0]$$

$$\Rightarrow \cot^{-1}(f(x)) = \tan^{-1}\left(\frac{1}{f(x)}\right) = \tan^{-1}(f(-x)) \text{ Ans.}$$

Q.4 (D)

$$I_n = \int_{-n}^n f(\{x\}) dx = \int_{-n}^n \frac{3\{x\}-2}{\{x\}+1} dx = 2n \int_0^1 \frac{3x-2}{x+1} dx = 2n \left(\int_0^1 \left(3 - \frac{5}{x+1} \right) dx \right) = 2n (3x - 5 \ln(x+1)) \Big|_0^1$$

$$\Rightarrow 2n (3 - 5 \ln 2) = 8 \ln \left(\frac{e^3}{32} \right) \Rightarrow 2n \ln \left(\frac{e^3}{32} \right) = 8 \ln \left(\frac{e^3}{32} \right) \Rightarrow n = 4$$

Q.5 (B)

$$I = \int_{-\frac{3}{2}}^{\frac{5}{2}} f(x) dx \quad \dots(1)$$



$$I = \int_{-\frac{3}{2}}^{\frac{5}{2}} f(1-x) dx \quad \dots(2)$$

$$2I = \int_{-\frac{3}{2}}^{\frac{5}{2}} ((f(x) + f(1-x))) dx$$

$$\text{Now, } f\left(\frac{x}{x-1}\right) + f\left(\frac{1}{1-x}\right) = 4 \Rightarrow f(t) + f(1-t) = 4$$

$$\therefore 2I = 4 \cdot \left(\frac{5}{2} + \frac{3}{2}\right) \Rightarrow I = 8. \text{ Ans.}$$

Q.6 (D)

$$\lim_{x \rightarrow 0} \left(\frac{\sin x + \sin 2x + \dots + \sin px}{2x} \right) = 5$$

$$\frac{(1+2+\dots+p)}{2} = 5 \Rightarrow p(p+1) = 20 \Rightarrow p = 4$$

Q.7 (D)

$$\text{Let } 4x = u \text{ and } 2t = v$$

$$\Rightarrow dx = \frac{du}{4} \text{ and } dt = \frac{dv}{2}$$

$$\therefore \int_0^4 \frac{u}{4} f(u) \frac{du}{4} + \int_0^4 \frac{v}{2} f(v) \frac{dv}{2} = \frac{5}{\lambda} \int_0^4 z f(z) dz$$

$$\Rightarrow \frac{1}{16} + \frac{1}{4} = \frac{5}{\lambda} \Rightarrow \frac{5}{16} = \frac{5}{\lambda} \Rightarrow \lambda = 16 \text{ Ans.}$$

Q.8 (D)

$$\text{Let } I = \int_1^4 \frac{3}{x} e^{\sin(x^3)} dx ; \text{ Put } x^3 = t$$

$$= \int_1^{64} \frac{e^{\sin t}}{t} dt = F(64) - F(1) \Rightarrow K = 64 \text{ Ans.}$$

Q.9 (B)

$$J = \int_0^1 \frac{\sin x}{\sqrt{x}} dx < \int_0^1 \frac{x}{\sqrt{x}} dx = \int_0^1 \sqrt{x} dx = \frac{2}{3} \left(x^{\frac{3}{2}} \right)_0^1 = \frac{2}{3}$$



$$\text{Also, } K = \int_0^1 \frac{\cos x}{\sqrt{x}} < \int_0^1 \frac{dx}{\sqrt{x}} = (2\sqrt{x})_0^1 = 2 \Rightarrow K < 2.$$

Q.10 (A)

$$\text{Put } x = t^3 + 2t - 1$$

$$\int_{11}^{71} \frac{dx}{3(f(x))^3 - 2f(x)} = \int_2^4 \frac{(3t^2 + 2)dt}{3t^3 - 2t}$$

$$\therefore I = \int_2^4 \frac{(3t^2 + 2)dt}{3t^3 - 2t} = \int_2^4 \frac{\left(3 + \frac{2}{t^2}\right)}{\left(3t - \frac{2}{t}\right)} dt$$

$$= \left(\ln \left(3t - \frac{2}{t} \right) \right)_2^4 = \ln \left(\frac{23}{2} \right) - \ln(5) \Rightarrow \ln \left(\frac{23}{10} \right)$$

$$\therefore p - q = 13 \text{ Ans.}$$

Q.11 (A)

Q.12 (B)

$$x^4 - 4x + 3 = x^4 - f'(1)x^3 + f''(1)x^2 - 4x + f(1)$$

which is a cubic equation and it has 4 distinct roots

$\therefore$ It is an identity

$$f'(1) = 0, f''(1) = 0, f(1) = 3$$

$$\therefore f(x) = (x-1)^3 + 3$$

(i) $\lim_{x \rightarrow 1} \frac{(x-1)^3 - \sin^3(x-1)}{(x-1)^5}$

$$x - 1 = t$$

$$\lim_{t \rightarrow 0} \frac{t^3 - \sin^3 t}{t^5} = \lim_{t \rightarrow 0} \left(\frac{t - \sin t}{t^3} \right) \left(\frac{t^2 + t \sin t + \sin^2 t}{t^2} \right) = \frac{1}{6} \times 3 = \frac{1}{2}$$

(ii) $\int \frac{dx}{(x-1)^2 + 4} = \frac{1}{2} \tan^{-1} \left(\frac{x-1}{2} \right) + c$

$$\therefore g(x) = \tan^{-1} \left(\frac{x-1}{2} \right)$$

$$\Rightarrow g(5) + g(7) = (\tan^{-1} 2 + \tan^{-1} 3) = \frac{3\pi}{4}.$$

Q.13 (C)

Q.14 (A)

$$\ln(f(x+y)) = \ln(f(x)) + \ln(f(y)) + 2xy - 3$$

Partially differentiating w.r.t. x



$$\frac{f'(x+y)}{f(x+y)} = \frac{f'(x)}{f(x)} + 2y$$

Putting $x=0$, $\frac{f'(y)}{f(y)} = \frac{f'(0)}{f(0)} + 2y$

$$\Rightarrow \frac{f'(y)}{f(y)} = 2 + 2y \Rightarrow \ln f(y) = 2y + y^2 + c$$

$$y=0, f(0) = e^3, c = 3$$

$$\therefore f(y) = e^{y^2+2y+3}$$

(i) $\frac{f'(x)}{f(x)} = \frac{e^{x^2+2x+3}}{e^{x^2+2x+3}} \cdot (2x+2) \Rightarrow \frac{f'(2)}{f(2)} = 6$

(ii) $f(x) = e^{x^2+2x+3} = y$
 $x^2 + 2x + 3 = \ln y$

$$(x+1)^2 = \ln y - 2 \Rightarrow x = -1 \pm \sqrt{\ln y - 2}$$

$$\text{For } x \in [0, 1] \rightarrow y \in [e^3, e^6]$$

$$\therefore x = -1 + \sqrt{\ln y - 2} \Rightarrow f^{-1}(x) = -1 + \sqrt{\ln x - 2}$$

$$\therefore \int_0^1 f(x) dx + \int_{e^3}^{e^6} (-1 + \sqrt{\ln x - 2}) dx + \int_{e^3}^{e^6} -1 \cdot dx \Rightarrow e^6 - (e^6 - e^3) = e^3$$

Q.15 (BD)

$$\text{Let } \frac{1-\sqrt{x}}{1+\sqrt{x}} = t \Rightarrow \frac{(1+\sqrt{x})\left(\frac{-1}{2\sqrt{x}}\right) - \frac{1}{2\sqrt{x}}(1-\sqrt{x})}{(1+\sqrt{x})^2} \Rightarrow \frac{-2}{2\sqrt{x}(1+\sqrt{x})^2} = dt$$

$$\text{Now, } I = -\int \sin 2t dt = \frac{\cos 2t}{2} + c = \frac{2\cos^2 t - 1}{2} + c$$

$$\Rightarrow \cos^2 t + k \Rightarrow \left\{ \cos\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right) \right\}^2 + k \Rightarrow f(x) = \cos x \quad \text{Ans.}$$

Q.16 (AD)

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx = \frac{4ab}{\sqrt{3}}$$

$$\Rightarrow \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi dx}{2 - \cos\left(x + \frac{\pi}{3}\right)} + 0 = \frac{4ab}{\sqrt{3}} \Rightarrow 2\pi \int_0^{\frac{\pi}{3}} \frac{dx}{2 - \cos\left(x + \frac{\pi}{3}\right)} = \frac{4ab}{\sqrt{3}}$$

$$\Rightarrow \frac{4\pi}{\sqrt{3}} \tan^{-1} \frac{1}{2} = \frac{4ab}{\sqrt{3}}$$

Q.17 (ABC)

$$I = \int \frac{2\sqrt{x} dx}{2\sqrt{x}(1+\sqrt{x})^{2010}};$$

$$\text{Put } 1 + \sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}} dx = dt$$

$$= 2 \int \frac{(t-1)dt}{t^{2010}} = 2 \int (t^{-2009} - t^{-2010}) dt = 2 \left(\frac{t^{-2008}}{-2008} - \frac{t^{-2009}}{-2009} \right) + C$$

$$= \frac{2}{2009(1+\sqrt{x})^{2009}} - \frac{2}{2008(1+\sqrt{x})^{2008}} + C$$

$$\therefore \alpha = 2009, \beta = 2008$$

Q.18 (A) S ; (B) Q ; (C) P, S ; (D) R

$$\int \frac{(\sin x + \cos x) + x(\sin x - \cos x)}{(1 + \sin 2x)} dx$$

$$= \int \frac{(\sin x + \cos x) - x(\cos x - \sin x)}{(\sin x + \cos x)^2} dx = \int \frac{d}{dx} \left(\frac{x}{\sin x + \cos x} \right) = \frac{x}{\sin x + \cos x} + c$$

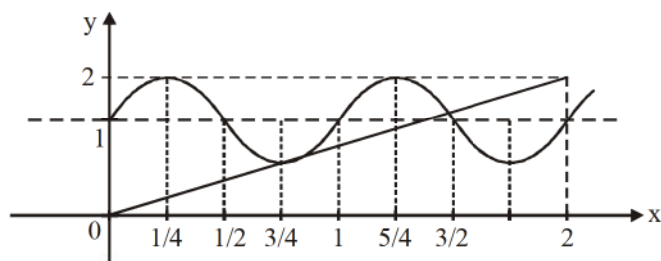
$$\Rightarrow f(x) = x, g(x) = \sin x + \cos x$$

$$(A) \int_{-2}^2 [x] dx = \int_{-2}^{-1} -2 dx + \int_{-1}^0 -1 dx + \int_0^1 0 dx + \int_1^2 1 dx = -2 - 1 + 0 + 1 = -2$$

$$(B) \lim_{x \rightarrow \frac{-\pi}{4}} \left(\frac{\sin x + \cos x}{x + \frac{\pi}{4}} \right) = \lim_{x \rightarrow \frac{-\pi}{4}} \frac{\cos x - \sin x}{1} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2} \quad [\text{L'Hospital Rule}]$$

$$(C) \int_{\pi/4}^{3\pi/4} (\sin x + \cos x) dx = (-\cos x + \sin x) \Big|_{\pi/4}^{3\pi/4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$(D) (g(\pi x))^2 = f(x) \Rightarrow 1 + \sin 2\pi x = x, \text{ number of solution} = 3$$



Q.19 [2]

$$\begin{aligned}\int_0^2 x f(x) dx &= \left(f(x) \cdot \frac{x^2}{2} \right)_0^2 - \int_0^2 f'(x) \cdot \frac{x^2}{2} dx = 2 f(2) - \frac{1}{2} \int_0^2 x^2 \cdot \frac{5x^3 - 6}{\sqrt{x^7 - 3x^4 + 4x^2}} dx \\ &= 2\sqrt{6} - \frac{1}{2} \int_0^2 \frac{5x^4 - 6x}{\sqrt{x^5 - 3x^2 + 4}} dx = 2\sqrt{6} - \frac{1}{2} \cdot 2 \cdot \left(\sqrt{x^5 - 3x^2 + 4} \right)_0^2 \\ &= 2\sqrt{6} - (2\sqrt{6} - 2) = 2 \text{ Ans.}\end{aligned}$$

Q.20 [2]

$$D = 4(f'(x))^2 - 4f(x)f''(x) = 0$$

$$\Rightarrow (f'(x))^2 = f(x)f''(x)$$

$$\Rightarrow \int \frac{f''(x)}{f'(x)} dx = \int \frac{f'(x)}{f(x)} dx$$

$$\Rightarrow \ln f'(x) = \ln f(x) + \ln c$$

$$f'(x) = c f(x)$$

$$\text{Putting } x=0, c=2$$

$$\therefore f'(x) = 2f(x) \Rightarrow \int \frac{f'(x)}{f(x)} dx = \int 2 dx$$

$$\Rightarrow \ln f(x) = 2x + c$$

$$\text{Putting } x=0 \Rightarrow c=0$$

$$\therefore \ln f(x) = 2x \Rightarrow f(x) = e^{2x}$$

$$\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1}{2x} \right)^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{e^{2x} - 1}{2x} - 1 \right)} = e^{\lim_{x \rightarrow 0} \left(\frac{e^{2x} - 1 - 2x}{2x^2} \right)} = e$$

$$\therefore [e] = 2 \text{ Ans.}$$